

Scalability & Future Plan

Date	29 June 2026
Team ID	
Project Name	VBCUA
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Local Streamlit Deployment	Supports limited concurrent users and is not suitable for production-scale access.	High
2	Limited Reference Concepts & AI Models	The application currently evaluates only predefined concepts and depends on fixed AI models, limiting flexibility and scalability.	Medium
3	Basic Security & User Management	No user authentication, role-based access control, or encrypted storage, limiting production readiness.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports individual users on a local Streamlit application.	Deploy on Streamlit Community Cloud, Render, or AWS with load balancing to support multiple concurrent users.
2	Data Storage	Processes uploaded audio files temporarily using local storage; no persistent database.	Integrate a cloud database (e.g., MongoDB Atlas or PostgreSQL) to store user profiles, transcripts, and evaluation history.
3	Performance	CPU-based processing with basic Streamlit caching for transcripts and audio features.	Enable GPU acceleration for Whisper, optimize AI inference, and implement advanced caching and asynchronous processing.
4	Security	Basic input validation; no authentication or	Implement user authentication, role-based

		encrypted communication.	access control, HTTPS, secure file uploads, and encrypted data storage.
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Future Roadmap:

Phase	Enhancement	Timeline	Expected Benefit
Phase 1	Cloud Deployment (Streamlit Community Cloud / Render / AWS)	3 Months	Enables public access, supports more concurrent users, and improves scalability.
Phase 2	Multilingual Speech Support & User Dashboard	6 Months	Allows users to explain concepts in multiple languages and track their learning progress through personalized dashboards.
Phase 3	AI Feedback Enhancement & Learning Management System (LMS) Integration	9 Months	Provides more detailed AI-generated feedback, personalized improvement suggestions, and seamless integration with educational platforms like Moodle or Google Classroom.
Phase 4	Mobile Application & Real-Time Progress Analytics	12 Months	Offers on-the-go access for students and educators, along with analytics to monitor learning trends and performance over time.